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SENIOR FULL STACK SOFTWARE ENGINEER

C#/.NET | Angular | AI-Enabled Engineering | SaaS | Cloud Architecture

PROFESSIONAL PROFILE

Senior Full Stack Software Engineer with 12+ years of commercial experience building and supporting enterprise software across fintech, AI-enabled analytics, workflow-heavy platforms and complex optimisation systems.

Strong commercial background in C#/.NET, Angular, React, and cloud-based application development, with experience spanning architecture, API design, distributed systems, CI/CD, testing, production support and technical leadership.

Most recently designed and delivered production AI capabilities within an established enterprise MIS platform, including a parent-embedded AI assistant, real-time streaming AI chat, MCP-based workflow orchestration and a natural-language analytics pipeline translating user requests into controlled SQL-driven outputs. Built the surrounding production safeguards including JWT validation, tenant/domain scoping, deterministic workflow checkpoints, prompt constraints, SQL guardrails, cancellation, timeout handling and structured error flows.

Experienced in working across the complete engineering lifecycle—from understanding product requirements and designing maintainable architectures through implementation, testing, deployment, monitoring and production support. Comfortable balancing new technology adoption with the security, maintainability and operational requirements of established SaaS products.

TECHNICAL SKILLS

Backend & APIs: C#, .NET 6/8, .NET Core, ASP.NET Core, REST APIs, Entity Framework Core, Java, Python

AI & Intelligent Systems: Azure OpenAI, Anthropic Claude API, Semantic Kernel, MCP, prompt engineering, natural-language-to-SQL workflows, AI-assisted analytics, Google OR-Tools

Frontend: Angular, TypeScript, RxJS, NgRx, React, Redux, HTML5, SCSS, CSS

Cloud & DevOps: Azure, AWS, GCP integrations, Azure DevOps, Docker, Azure Functions, Azure Service Bus, Bicep, CI/CD

Data: SQL Server, Azure SQL, MongoDB, Cosmos DB

Architecture & Engineering: Microservices, distributed systems, SaaS applications, SOLID, DDD, design patterns, TDD, CI/CD, API architecture, secure system integration

Production Engineering: SSE, JWT authentication/authorisation, monitoring, error handling, timeout/cancellation handling, code reviews, production troubleshooting

PROFESSIONAL EXPERIENCE

Senior Full-Stack Software Engineer | Bromcom, London

Oct 2024 – Present

Production AI Assistant & AI Workflow Engineering

- Architected and delivered a production-grade, parent-embedded AI assistant for Bromcom's school MIS platform, spanning a React user experience, .NET API gateway, MCP workflow service and Analytics on Demand microservice.
- Took responsibility across the AI feature lifecycle rather than treating AI as an isolated API integration—covering application architecture, frontend integration, authentication and context propagation, AI workflow orchestration, real-time streaming, natural-language analytics and production safeguards.
- Built a secure host-to-iframe integration using defined postMessage contracts to bootstrap authentication, propagate school and user context and synchronise application state between the established MIS product and the embedded AI experience.
- Developed the .NET API gateway responsible for validating JWTs, resolving claim-based school and domain scope, minting downstream tokens and normalising streamed responses between AI services and the frontend.
- Implemented real-time AI chat using Server-Sent Events (SSE), including status events, cancellation handling and timeout protection so long-running AI operations could be surfaced and controlled cleanly by the application rather than leaving users with blocking requests.
- Designed an MCP tool-server around a deterministic report-building state machine. Introduced structured Action Required checkpoints and strict tool-sequencing rules so AI-driven workflows could be constrained by application logic instead of relying solely on unconstrained model behaviour.
- Built an Analytics on Demand orchestration pipeline that converts natural-language requests into SQL-driven analytics and executive-level HTML outputs, connecting conversational AI with existing enterprise data.
- Applied layered prompt engineering and validation rules around the pipeline, including mode gating, cross-domain restrictions, date-filter gating, SQL safety guardrails and structured tool-to-UI protocols.
- Designed the AI integration with production security boundaries in mind, using JWT validation and claim-based school/domain scope resolution to prevent requests from operating outside the authorised application context.
- Combined deterministic application logic with AI-generated behaviour so critical workflow and data-access decisions remained controlled by software rules while AI was used where natural-language interpretation and generation added value.

- Built failure and resilience considerations into the streaming architecture through cancellation handling, timeout protection and controlled response proxying, helping make AI functionality suitable for an established production platform rather than a standalone prototype.

C#/.NET, Architecture & SaaS Engineering

- Designed and developed scalable C#/.NET REST APIs and reactive Angular applications supporting high-frequency scheduling and enterprise timetable workflows.
- Designed automated timetable generation using Google OR-Tools, processing thousands of scheduling permutations across teacher availability, room allocation, subject and band constraints.
- Translated complex scheduling rules into maintainable application and optimisation logic, balancing algorithmic capability with the operational requirements of an established enterprise platform.
- Worked across backend services, Angular/React interfaces, APIs and distributed components, providing full-stack ownership rather than working within a single technical layer.

Quality, Delivery & Production Support

- Built and supported cloud-native CI/CD delivery pipelines with automated monitoring.
 - Diagnosed and resolved production and platform bottlenecks across application and integration layers.
 - Contributed to architectural decisions, technical reviews and engineering standards while mentoring engineers through code reviews and maintainable implementation approaches.
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Team Leader / Senior Full-Stack Developer | MGT Fintec Synergies Ltd, London

Dec 2022 – Sep 2024

- Led a cross-functional engineering team delivering fintech applications and enterprise full-stack solutions within a financial technology environment.
 - Architected and developed scalable .NET Core APIs and Angular applications supporting financial data processing platforms.
 - Combined hands-on C#/.NET and Angular engineering with team leadership, architecture decisions, delivery ownership and stakeholder collaboration.
 - Worked with product owners and business stakeholders to translate financial and operational requirements into scalable technical solutions.
 - Introduced TDD and clean-code practices to improve maintainability and reduce regression-related issues across the applications delivered by the team.
 - Managed Azure infrastructure, deployment pipelines and production release processes, providing engineering ownership beyond application development alone.
 - Supported the delivery lifecycle from business requirements and technical design through development, testing, deployment and production operation.
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Senior Front-End Engineer | BlackSwan Technologies

Jul 2021 – Dec 2022

- Developed reusable Angular service libraries supporting authentication workflows across Azure, AWS and GCP environments.
 - Designed scalable Angular architectures and reusable UI components for enterprise applications using object-oriented engineering principles.
 - Implemented Azure Service Bus messaging patterns supporting asynchronous communication across distributed systems.
 - Built Azure DevOps CI/CD pipelines to improve repeatability of application deployment and release management.
 - Contributed beyond the frontend to backend microservices and AI-driven analytics integrations using Java and Python, working across distributed application boundaries.
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Senior Front-End Engineer | InEight

Feb 2020 – Jun 2021

- Led migration of established enterprise applications from JavaScript/Kendo UI to React, modernising the frontend while improving maintainability and user experience.
 - Implemented Azure Functions and cloud integrations supporting scalable frontend and serverless application workflows.
 - Developed CI/CD pipelines and infrastructure-as-code deployments using Azure DevOps and Bicep.
 - Improved production visibility and performance monitoring using Azure Application Insights, supporting faster diagnosis of application and operational issues.
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Senior Software Engineer | QualitApps

Nov 2018 – Jan 2020

- Developed scalable .NET Core APIs and Angular applications for enterprise web platforms.
 - Worked commercially across both backend and frontend engineering, building APIs, business functionality and modern web interfaces.
 - Integrated third-party APIs, authentication services and cloud-hosted systems across Azure, AWS and GCP.
 - Applied SOLID principles and established design patterns to improve maintainability and scalability as applications evolved.
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Full-Stack Team Leader / Software Engineer | Siyobit Holdings

2014 – 2018

- Built Angular and .NET applications supporting real-time financial and trading platforms.
- Developed scalable RESTful APIs and reusable frontend components for business-critical applications.
- Combined hands-on full-stack development with team leadership responsibilities.

- Supported CI/CD deployments and contributed to the migration of desktop WPF applications to modern web platforms.
 - Worked with established financial software where reliability, maintainability and controlled application evolution were important engineering considerations.
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EDUCATION

Bachelor of Information Technology

University of Moratuwa

CORE VALUE FOR SENIOR ENGINEERING ROLES

Production AI Engineering

Commercial experience integrating AI into an established enterprise application—not simply using AI coding assistants—including MCP workflows, natural-language analytics, streaming AI interactions, prompt constraints and controlled data access.

Strong Microsoft Engineering Foundation

12+ years of software engineering experience with significant commercial C#/.NET, Angular, SQL Server, Azure and enterprise application development.

Full-Stack Ownership

Comfortable moving between C#/.NET services, APIs, Angular/React interfaces, cloud infrastructure, authentication, distributed integrations and production troubleshooting.

Safe AI Integration

Experience combining probabilistic AI behaviour with deterministic application controls including workflow checkpoints, domain restrictions, JWT-based scope enforcement, SQL guardrails, cancellation and timeout handling.

Engineering Quality

Practical experience with TDD, SOLID, design patterns, code reviews, CI/CD, monitoring, production support and maintainability-focused architecture.

Technical Leadership

Previous Team Leader experience alongside current senior engineering responsibilities, with experience guiding architectural decisions, reviewing code, mentoring engineers and translating product requirements into deliverable technical solutions.

REFERENCES

Professional and character references available upon request.